FORTH ORIGINAL 1130

```
PAGE    1

// JOB          3333

LOG DRIVE    CART SPEC    CART AVAIL   PHY DRIVE
  0000         0001         0001         0000
  0001         3333         3333         0001

// ASM
*LIST ALL
                                ABS          FORTH
03C0                            ORG          /900
0900 0  0000            CONVE   DC           CONVERT EBCDIC TO FORTH
0901 0  C1FF                    LD     1 -1    CHARACTER SET
0902 0  D002                    STO    CV1+1
0903 0  63C1                    LDX    3 -63
0904 00 C7000000        CV1     LD     L3
0906 0  9100                    S      1     A
0907 00 4C18090B                BZ           CV2
0909 0  7301                    MDX    3 1
090A 0  70F9                    B            CV1
090B 0  733F            CV2     MDX    3 63
090C 0  1000                    NOP
090D 0  6B5F                    STX    3 TEMP
090E 0  C05E                    LD           TEMP
090F 0  D100                    STO    1     A
0910 00 4C800900                B      I     CONVE
0912 0  0000            ACCEP   DC           NEXT CHARACTER FROM DISK
0913 0  C1FD                    LD     1 -3    C
0914 00 94000B28                S      L     C2
0916 00 44180B2D                BSI    L     RECOR,+-    NEXT RECORD
0918 0  C1FD                    LD     1 -3    C
0919 0  4054                    BSI          FETCH
091A 0  40E5                    BSI          CONVE
091B 00 4C800912                B      I     ACCEP
091D 0  0000            RETRV   DC           FROM CORE
091E 0  C1FD                    LD     1 -3    C
091F 0  4C4E                    BSI          FETCH
0920 00 4C80091D                B      I     RETRV
0922 0  0000            NEXT    DC           NEXT WORD
0923 0  C1FA                    LD     1 -6    W1
0924 0  D102                    STO    1 2     W
0925 0  C03A                    LD           BL2
0926 0  D104                    STO    1 4     WORD+1
0927 00 4580FFFE        NE1     BSI    I1 -2   ACCEPT
0929 0  C100                    LD     1     A
092A 0  9036                    S            BL
092B 00 4C200931                BNZ          NE2
092D 0  C1FD                    LD     1 -3    C
092E 0  803D                    A            ONE
092F 0  D1FD                    STO    1 -3    C
0930 0  70F6                    B            NE1
0931 0  902B            NE2     S            SC
0932 00 4C08093A                BNP          ALPHA
```

```
FO.FO
  1
  2
  3
  4
  5
  6
  7
  8
  9 .DEP  LOC DEPOSIT,  .DEPOSIT  IC INC=,
 10 .INST OR DEPOSIT,
  1
  2 .LD C000,   .ST D000,
  3 .ADD 8000,   .SUB 9000,
  4 .MUL  A000,   .DIV A800,
  5 .LX 6000,  .SX 6800,  .MX 7000,
  6 .B 7000,   .BL 4800,   .BSI 4000,
  7 .XIO 800,
  8 .X1  100 OR,   .X2  200 OR,   .X3  300 OR,
  9
 20 .MDM  7400 INST DEPOSIT,
  1 .ZERO 1810 DEPOSIT,
  2 .XCH 18D0 DEPOSIT,
  3 .I 480 INST DEPOSIT,   .L 400 INST,
  4 .LONG  L DEPOSIT,
  5 .STORE  ST LONG,  .LOAD  LD LONG,  .CALL BSI LONG,
  6 .ACC  1 B INST  DEPOSIT  FE INST,
  7
  8 .GDEP  IC 1 MDM  IC ST I,
  9 .GOR  E800,  .GAND  E000,  .GWAIT  3000 DEPOSIT,
 30 .REL  X3 LONG,   .RELS  X3 INST,   .BASE  LX REL,
  1 .BASA  1 ST INST  0 BASE,
  2 .TOD  1880 INST,   .TOM  1000 INST,
  3 .TXD  1880 INST,   .TXM  1080 INST,
  4 .TAD  1800 INST,
  5 .ZF  10A0 DEPOSIT,  .LF  ZF LOAD,
  6
  7 .FS  X2 INST,
  8 .LS  LD FS,   .SS ST FS,   .LSI  LD X2 I,   .SSI  ST X2
  9 .MS  MX FS,
 40 .RETURN  BL I,
  1
  2 OPERATION CONSTANT  E1 LX X3 I  LD LOC LIT ACC
  3    2 ST X3 INST  0 LS  3 ST X3 INST  RETURN
  4
  5
  6 OPERATION REVERT  E LOAD  E1 STORE  E1 FC MDM  RETURN
  7 OPERATION TOP  E1 LOAD  E STORE  E 4 MDM  RETURN
  8
  9 OPERATION DUP  0 LS  1 MS  0 SS  RETURN
 50 OPERATION VALUE  0 LSI  0 SS  RETURN
  1 OPERATION DROP  FF MS  RETURN
  2 OPERATION RAISE  1 MS  RETURN
  3 OPERATION SWAP  0 LS  XCH  FF LS  0 SS  XCH  FF  SS  RE
  4 .MSI  LD SWAP ACC  DUP  ADD FS  SS,
  5
  6 OPERATION +  0 LS  FF MS  0 ADD FS  0 SS  RETURN
  7 OPERATION AND  0 LS  FF MS  0 E000 FS 0 SS RETURN
  8 .DISP  FF AND,
  9 OPERATION -  FF MS 0 LS  1 SUB FS  0 SS  RETURN
```

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 60 OPERATION MINUS  ZERO  0 SUB FS  0 SS  RETURN
  1 OPERATION /  FF LS  1890 DEPOSIT  0 DIV FS  FF SS XCH 0 SS  RETURN
  2 OPERATION *  FF LS  0 MUL FS  XCH  FF MS  0 SS  RETURN
  3
  4 OPERATION PUSH  0 LSI  ADD 1 ACC  0 SSI  0 SS  FF LS  0 SSI
  5    FE MS  RETURN
  6 OPERATION PULL  0 LSI 1 SS SUB 1 ACC 0 SSI 1 LSI 0 SS RETURN
  7 .STACK  1 MS  LOAD  0 SS,
  8 .LITS  LD SWAP ACC  1 MS  0 SS,
  9 OPERATION VECTOR  0 LS  GDEP  IC STACK  LOC CONSTANT CALL
 70    FF ADD FS  IC STORE  FE MS  RETURN
  1 .XS  4 VECTOR, XS  XS XS=
  2 .TRANSIENT  IC INC  XS PUSH,
  3 OPERATION EXECUTE  LD BL 480 OR  ACC GDEP  XS LD I  GDEP
  4    XS LX X3 I  0 BSI X3 I  XS LITS  LOC PULL CALL
  5    SUB 1 ACC  IC STORE  FF MS  RETURN
  6
  7 .BACK  IC VALUE - DISP B INST,
  8 .MARK  IC VALUE  SWAP  LX X3 INST,
  9 .LOOP  FF MX X3 INST  BACK,
 80 .START  2 B INST  DEPOSIT  IC INC
  1    FD LD INST  FD ST INST,
  2 .STOP  DUP FF MDM  1+ BACK,
  3
  4 .CONDITION  0 LS  FF MS  BSI OR L  LOC NEXT DEPOSIT  RETURN,
  5 OPERATION NONZERO  18 CONDITION
  6 OPERATION FALSE  20 CONDITION
  7 OPERATION EVEN  04 CONDITION
  8 .IF  BL 400 OR INST  IC INC,  .THEN  IC VALUE 1+  SWAP=,
  9 .ELSE  IC VALUE 3+  SWAP=  0 IF,
 90 .POSITIVE  8,  .NEGATIVE  10,  .EQUAL  20,  .NOT  18,
  1
  2 .PROGRAM  1 B INST  0 DEPOSIT,
  3 OPERATION QUEUE  1- PROGRAM  GWAIT BACK
  4 .ENQUEUE  LOAD  LOC QUEUE 2+ STORE
  5    LD  1 B OR  ACC  LOC QUEUE 1+ STORE  LOC QUEUE 1+ BL LONG,
  6 .REACTIVATE  LD  BL 400 OR  ACC  LOC QUEUE 1+ STORE,
  7 .BNZ  BL 20 OR,
  8
  9 .BUFFER  DUP DUP DUP DUP VALUE 2/ DROP+ 2+
100    SWAP 1+= 2+ SWAP=,
  1 OPERATION EXCHANGE  0 LS  ADD 1 ACC  1 SS  0 LSI
  2    XCH  1 LSI  0 SSI  XCH  1 SSI  FF MS  RETURN
  3 .CLEAR  LOAD  4840 I,
  4 .NOP  B DEPOSIT,
  5 .ALIGN  IC VALUE  EVEN NOP,
  6 .GXIO  ALIGN  1 XIO INST  2 B INST  DEPOSIT DEPOSIT,
  7 .WTC  1+ 2*,
  8
  9 OPERATION FILL  0 LS  FF MS  1 ST INST  0 START  LOC NEXT CALL
110    LOC HEX CALL  0 LS  GDEP  FF MS  STOP  RETURN
  1 .DATA  IC VALUE 1+ CONSTANT  DROP,
  2 .CSKB 40 DATA FILL 2000 1000 800 400 200 100 80 40 20 10
```

```
      3 9000 8800 8400 8200 8100 8080 8040 8020 8010 5000 4800 4400 4200 4100
      4 4080 4040 4020 4010 2800 2400 2200 2100 2080 2040 2020 2010  0000
      5 8820 0420 8220 8120 80A0 8060 8000 4820 4420 4220 4120 40A0 4060
      6 4000 3000 2420 2220 2120 20A0 2060 0820 8420 0220 0120 00A0 0060
      7 0000, CSKB
      8 .CSCP 40 DATA FILL C400 FC00 D800 DC00 F000 F400 D000 D400 E400 E000
      9 3C00 1800 1C00 3000 3400 1000 1400 2400 2000 7C00 5800 5C00 7000
    120 7400 5000 5400 6400 6000 9800 9C00 B000 B400 9000 9400 A400 A000
      1 2100 0200 C000 DE00 FE00 DA00 C600 4400 4200 4000 D600 F600 D200
      2 F200 8400 BC00 8000 0600 BD00 4600 8600 8200 0000 0400 E600 C200
      3 E200 8600, CSCP
      4 .A  2 VECTOR, A
      5 .RESTORE  A 2+ STORE  0F01 0 GXIO,
      6 .CONTINUE  A 2+  CLEAR,
      7 OPERATION READY  RESTORE  REACTIVATE  CONTINUE
      8 .CHA  A 1+ STORE  0900 A 1+ GXIO  LD LOC READY ACC  0C STORE,
      9 OPERATION CHARACTER  RESTORE  0A00 A 1+ GXIO  A 1+ LOAD
    130    ST X1 DEPOSIT  LOC CONVERT CALL  CSCP LD REL CHA  CONTINUE
      1 OPERATION ACCEPT  LD LOC CHARACTER ACC 0C STORE 0C00 0 GXIO ENQUEUE
      2 OPERATION CONSOLE  LD CSKB 3F+ ACC  FF ST X1 INST  LD LOC ACCEPT ACC
      3    FE ST X1 INST  RETURN
      4
      5 OPERATION TYP  CHA ENQUEUE
      6 .RED  900,  .BLACK  500,
      7 OPERATION RIBBON  0 LS  FF MS  LOC TYP CALL LD 8100 ACC
      8    LOC TYP CALL  RETURN
      9 OPERATION TYPE
    140    0 LS  1 ST INST  0 START  FF LS  LOC FETCH CALL
      1       BASA  SUB 3F ACC  POSITIVE IF  3F BASE  THEN
      2    CSCP LD REL  LOC TYP CALL  FF 1 MSI STOP FE MS RETURN
      3 .MESSAGE  BLACK RIBBON  TYPE  RED RIBBON,
      4 .ALPHA  14 VECTOR, ALPHA
      5 .REPLY  ALPHA WTC  ALPHA VALUE  MESSAGE,
      6
      7 .D  3 VECTOR, D
      8 .QUERY  8F01 0 GXIO,
      9 OPERATION DONE  D STORE  QUERY  REACTIVATE  D CLEAR
    150 OPERATION HOME  ZERO  D 1+ ST LONG  QUERY  GAND 0800 ACC
      1    DUP BNZ I  8C04 C8 GXIO  ENQUEUE
      2 .IN  154 VECTOR, IN  140 IN=
      3 .BZ  BL 18 OR,        141
      4
      5 .M  LD,  .CS  MS,  .ZLT  ZERO,
      6 .TS  SS,  .TM  STORE,  .TR  ST INST,  .TA  ST REL,
      7 .ST  FS,  .MT  LONG,  .RT  INST,  .AT  REL,  .LT  SWAP ACC,
      8
      9 .FIXUP  6 M RT  3 TAD  3 TXM  3 TR,
    160 OPERATION CYLINDER  3 TOD  2 TS  D 1+ SUB MT  DUP BZ I
      1    POSITIVE IF  3 TOM  3 TXD
      2    ELSE 1 TS  10 TXD  ZLT  1 SUB ST
      3    THEN  ALIGN NOP  6 TR
      4    FIXUP  8C00 0 GXIO  2 M ST  D 1+ TM
      5    8 M LT  D 2+  TM  ENQUEUE
      6 OPERATION SECTOR  0 M ST  LOC CYLINDER CALL 0 M ST  FF CS
      7    7 GAND LT  1 TS  D 2+ SUB MT  DUP BZ I  1 M ST  D 2+ TM
      8    3 TXD  ALIGN  FIXUP  8E00 IN GXIO  ENQUEUE
      9 OPERATION WRITE  D 2+ M MT 3 TXD ALIGN FIXUP  8D00 IN GXIO  ENQUEUE
    170 .CHARACTER  FFFE BSI X1 I  FD M X1 RT  1 ADD LT  FD X1 TR,
      1 .REP  FE M ST  2 TR  0 M X1 RT  3C SUB LT  SWAP DUP BZ I SWAP
      2    0 M ST  DEP CALL  0 1 MSI,
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  3 OPERATION FILL  28 START  CHARACTER  REP  STOP
  4    8100 M LT  LOC TYP CALL  RETURN
  5 OPERATION BLANK  0 M ST  1 TOD  FF CS  LX X3 GOR LT
  6    9 TR  0 M ST  1 TOD  1 SUB LT  5 TR  2424 M LT  0 MARK
  7       0 TA  LOOP  RETURN
  8 OPERATION LIZ  ALPHA WTC M LT  1 TS  2 TS
  9    28 M LT  3 TS  3 CS  LOC BLANK CALL
180    LOC FILL CALL  0 M ST  FF SUB ST  ALPHA TM  FD CS  RETURN
  1 OPERATION ' 3C M LT  1 TS  1 CS  LOC LIZ CALL  RETURN
  2 OPERATION (  2F M LT  1 TS  1 CS  LOC LIZ CALL  RETURN
  3 .$ 'OK' REPLY,
  4    .FILE  7 VECTOR, FILE  .LINE INTEGER,
  5    OPERATION POSITION  0 M ST  4 TOD  1 CS  0 TS
  6       LOC SECTOR CALL  0 M ST  F GAND LT  14 MUL LT  XCH
  7       IN ADD LT  0 TS RETURN
  8    OPERATION FILL  28 START  LOC ACCEPT CALL  REP  STOP  RETURN
  9    .L  LINE VALUE,  .RELATIVE  FILE 5+ VALUE+,
190 .T  DUP LINE=  RELATIVE POSITION WTC  28 MESSAGE,
  1 .EXAMINE  SWAP  IN WTC + SWAP  MESSAGE,
  2 .EMPLACE  3C  IN WTC+ 28 FILL  DROP DROP DROP  WRITE,
  3 .S INTEGER,  .SV S VALUE,
  4 .MOVE TRANSIENT  MARK  SWAP M AT  SWAP TA  LOOP  EXECUTE,
  5 .COMPARE  2 TR  MARK  0 M AT  SWAP  SUB AT EQUAL IF SWAP LOOP ELSE,
  6 OPERATION SEARCH  IN 6- M LT  S TM  IN 1+ M MT  1 TR  0 START
  7    S 7 MDM  S M MT  ALPHA 4 COMPARE  SWAP STOP THEN 1 CS
  8    0 TS  RETURN
  9 .CREATE  2E0 SECTOR  IN 1+ DUP INC 1- 7*+ S= SV VALUE 1+ SV 5+=
200    RAISE SV 6+=  RAISE F+ SV 7+=  SV ALPHA 4 MOVE WRITE,
  1    .WARN  'NO SUCH FILE' REPLY,
  2 .ACTIVATE  2E0 SECTOR  SEARCH NONZERO WARN  FILE SV 7 MOVE,
  3 .DELETE  2E0 SECTOR  0 SV 1+= WRITE,
  4
  5 .SEC INTEGER,
  6 OPERATION STRAIGHT  FD X1 M RT  IN 140+ WTC  SUB LT  EQUAL IF
  7    SEC 1 MDM  SEC M MT  1 CS  0 TS  LOC SECTOR CALL
  8     IN WTC M LT  FD X1 TR  THEN
  9    FD X1 M RT  LOC FETCH CALL  RETURN
210 OPERATION SET  LOC STRAIGHT M LT  FE X1 TR
  1    0 M ST  FF CS  FD X1 TR  RETURN
  2 .INTERPRET TOP  FILE 5+ VALUE 10/ 14* IN+ WTC
  3    SWAP DUP SEC=  SECTOR  SET,
  4 .RETRIEVE  ACTIVATE INTERPRET,
  5
  6 .XT  SX INST  0 M LT,
  7 OPERATION FORGET  1 X3 XT  E TM  4 SUB LT  E1 TM  3 X3 M RT
  8    IC TM  RETURN
  9 OPERATION REMEMBER  LOC ENTRY CALL
220    LOC FORGET M LT  2 X3 TR  IC M MT  3 X3 TR  RETURN
  1
  2 LOC DONE 0A=  HOME  'HI THERE' REPLY  CONSOLE
```

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PAGE    2

0934 0  902A                    S       THREE
0935 00 4C280954                BN      SPECI
0937 0  9026                    S       TWO
0938 00 4C200954                BNZ     SPECI
093A 0  C102            ALPHA   LD    1 2   W   ALPHABETIC
093B 0  4043                    BSI     DEPOS
093C 00 4580FFFC                BSI   I1 -4    SAVE
093E 0  C1FD                    LD    1 -3    C
093F 0  802C                    A       ONE
0940 0  D1FD                    STO   1 -3    C
0941 0  C102                    LD    1 2   W
0942 0  8029                    A       ONE
0943 0  D102                    STO   1 2
0944 00 4580FFFE                BSI   I1 -2    ACCEP
0946 0  C100                    LD    1    A
0947 0  9019                    S       BL
0948 00 4C18094D                BZ      AL1
094A 00 4CB00922                BP    I NEXT
094C 0  70ED                    B       ALPHA
094D 00 4580FFFC        AL1     BSI   I1 -4    SAVE
094F 0  C1FD                    LD    1 -3    C
0950 0  801B                    A       ONE
0951 0  D1FD                    STO   1 -3    C
0952 00 4C800922                B     I NEXT
0954 0  C102            SPECI   LD    1 2   W   SPECIAL CHARACTER
0955 0  4029                    BSI     DEPOS
0956 00 4580FFFC                BSI   I1 -4    SAVE
0958 0  C1FD                    LD    1 -3    C
0959 0  8012                    A       ONE
095A 0  D1FD                    STO   1 -3    C
095B 00 4C800922                B     I NEXT
095D 0  0014            SC      DC      20
095E 0  0002            TWO     DC      2
095F 0  0003            THREE   DC      3
0960 0  2424            BL2     DC      /2424
0961 0  0024            BL      DC      /24
0962 0  0000            SAVE    DC          BUILD MEMORY
0963 0  C1FB                    LD    1 -5    D
0964 0  8007                    A       ONE
0965 0  D1FB                    STO   1 -5    D
0966 0  4018                    BSI     DEPOS
0967 00 4C800962                B     I SAVE
0969 0  0000            SAVE0   DC          DUMMY
096A 00 4C800969                B     I SAVE0
096C 0  0001            ONE     DC      1
096D 0  0000            TEMP    DC      0
096E 0  0000            FETCH   DC      0    FETCH CHARACTER
096F 0  1881                    SRT     1
0970 0  D0FC                    STO     TEMP
0971 0  1081                    SLT     1
0972 00 4C040978                BOD     FE1
0974 00 C480096D                LD    I TEMP   EVEN CHARACTER
0976 0  1808                    SRA     8
0977 0  7003                    B       FE2
0978 00 C480096D        FE1     LD    I TEMP   ODD CHARACTER
097A 0  E003                    AND     FF
097B 0  D100            FE2     STO   1    A
097C 00 4C80096E                B     I FETCH
097E 0  00FF            FF      DC      /00FF
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PAGE   3

097F 0  0000          DEPOS DC          DEPOSIT CHARACTER
0980 0  1881                SRT     1
0981 0  D0EB                STO     TEMP
0982 0  1081                SLT     1
0983 00 4C04098A            BOD     DE1
0985 0  C100                LD    1   A   EVEN CHARACTER
0986 0  100A                SLA     .10
0987 0  1802                SRA     2
0988 0  E8D8                OR      BL
0989 0  7004                B       DE2
098A 00 C480096D    DE1     LD    I TEMP  ODD CHARACTER
098C 0  E005                AND     FF00
098D 0  E900                OR    1   A
098E 00 D480096D    DE2     STO   I TEMP
0990 00 4C80097F            B     I DEPOS
0992 0  FF00          FF00  DC        /FF00
0993 0  0000          DO    DC          INTERPRET WORD
0994 00 67800AD8            LDX   I3 E
0996 0  73FC          DO1   MDX   3 -4
0997 0  C300                LD    3
0998 00 4C980993            BZ    I DO
099A 0  9103                S     1 3   WORD
099B 00 4C200996            BNZ     DO1
099D 0  C301                LD    3 1
099E 0  9104                S     1 4   WORD+1
099F 00 4C200996            BNZ     DO1
09A1 00 4C800993            B     I DO
09A3 0  0000          UNDEF DC          UNDEFINED SYMBOL
09A4 0  C103                LD    1 3   WORD
09A5 0  E004                AND     F000
09A6 00 441809AB            BSI   L HEX,+-
09A8 00 4C8009A3            B     I UNDEF
09AA 0  F000          F000  DC        /F000
09AB 0  0000          HEX   DC          HEXADECIMAL LITERAL
09AC 0  C102                LD    1 2   W
09AD 0  91FA                S     1 -6  W1
09AE 0  D0BE                STO     TEMP
09AF 00 6780096D            LDX   I3 TEMP
09B1 0  7201                MDX   2 1
09B2 0  1810                SRA     16
09B3 0  D200                STO   2
09B4 0  C1FA                LD    1 -6  W1
09B5 0  D102          HE1   STO   1 2   W
09B6 0  40B7                BSI     FETCH
09B7 0  C200                LD    2
09B8 0  1004                SLA     4
09B9 0  E900                OR    1   A
09BA 0  D200                STO   2
09BB 0  C102                LD    1 2   W
09BC 0  80AF                A       ONE
09BD 0  73FF                MDX   3 -1
09BE 0  70F6                B       HE1
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PAGE   4

09BF 00 4C8009AB          B    I  HEX
09C1 0  0000        ENTRY DC        INITIAL ENTRY
09C2 00 44000922          BSI  L  NEXT
09C4 00 74040AD7          MDM  L  E1,4
09C6 00 67800AD7          LDX  I3 E1
09C8 00 6F000AD8          STX  L3 E
09CA 00 74040AD8          MDM  L  E,4
09CC 0  C103              LD   1 3   WORD
09CD 0  D300              STO  3
09CE 0  C104              LD   1 4   WORD+1
09CF 0  D301              STO  3 1
09D0 00 4C8009C1          B    I  ENTRY
09D2 0  0000        ENTER DC        ADD TO SYMBOL TABLE
09D3 0  40ED              BSI     ENTRY
09D4 00 C4000A55          LD   L  INTER
09D6 0  D302              STO  3 2
09D7 00 C4000D83          LD   L  IC
09D9 0  1001              SLA     1
09DA 0  8091              A       ONE
09DB 0  D1FB              STO  1 -5   D
09DC 0  808F              A       ONE
09DD 0  D303              STO  3 3
09DE 0  C016              LD      ASAVE   SKIP OVER DEFINITIO
09DF 0  D1FC              STO  1 -4   SAVE
09E0 00 44000922    EN1   BSI  L  NEXT
09E2 0  C015              LD      COMMA
09E3 00 94000AE2          S    L  WORD
09E5 00 4C2009EF          BNZ     EN2
09E7 0  C1FB        EN3   LD   1 -5   D
09E8 0  1801              SRA     1
09E9 00 D4000D83          STO  L  IC
09EB 0  C00A              LD      ASAV0
09EC 0  D1FC              STO  1 -4   SAVE
09ED 00 4C8009D2          B    I  ENTER
09EF 0  C007        EN2   LD      DOT
09F0 00 94000AE2          S    L  WORD
09F2 00 4C2009E0          BNZ     EN1
09F4 0  70F2              B       EN3
09F5 0  0962        ASAVE DC        SAVE
09F6 0  0969        ASAV0 DC        SAVE0
09F7 0  3024        DOT   DC        /3024   SEMICOLON
09F8 0  3424        COMMA DC        /3424   ,
09F9 0  0000        X3    DC
09FA 0  6BFE              STX  3 X3
09FB 00 66000AF7    START LDX  L2 STACK+1   RESTART LOCATION
09FD 00 65000ADF          LDX  L1 A
09FF 00 67000B07          LDX  L3 INTST
0A01 00 6F000B06          STX  L3 R
0A03 00 67000DC0          LDX  L3 E2
0A05 00 6F000AD7          STX  L3 E1
0A07 0  7304              MDX  3 4
0A08 00 6F000AD8          STX  L3 E
0A0A 00 67001860          LDX  L3 2*SECT+74
0A0C 00 6F000AF6          STX  L3 STACK
0A0E 00 6F000B28          STX  L3 C2
0A10 00 670000B3          LDX  L3 /B3
0A12 00 6F00000A          STX  L3 /A
0A14 00 67000A31          LDX  L3 FORTH
0A16 00 6F000AF7          STX  L3 STACK+1
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PAGE   5

0A18 00 67000912          LDX  L3 ACCEP
0A1A 00 6F000ADD          STX  L3 A-2
0A1C 00 67000C09          LDX  L3 BCD+63
0A1E 00 6F000ADE          STX  L3 A-1
0A20 00 67000969          LDX  L3 SAVE0
0A22 00 6F000ADB          STX  L3 A-4
0A24 00 670019FF          LDX  L3 /19FF
0A26 00 6F000D83          STX  L3 IC
0A28 00 670010EE          LDX  L3 /10EE
0A2A 00 6F000C0B          STX  L3 SECT
0A2C 0  400E              BSI     RECUR
0A2D 00 02053580          LINK    BALO
0A31 0  72FF        FORTH MDX   2 -1
0A32 0  C201              LD    2 1
0A33 0  D1FD              STO   1 -3   C
0A34 00 44000922    FO1   BSI  L  NEXT
0A36 00 44000993          BSI  L  DO
0A38 00 47800002          BSI  I3 2
0A3A 0  70F9              B       FO1
0A3B 0  0000        RECUR DC
0A3C 0  C0FE              LD      RECUR
0A3D 00 D4800B06          STO  I  R
0A3F 00 74010B06          MDM  L  R,1
0A41 0  72FF              MDX   2 -1
0A42 00 4E800001          B    I2 1
0A44 0  0000        RETUR DC
0A45 00 74FF0B06          MDM  L  R,-1
0A47 00 67800B06          LDX  I3 R
0A49 00 4F800000          B    I3 0
0A4B 0  0000        INC   DC          INCREMENT COUNTER
0A4C 00 C6800000          LD   I2
0A4E 00 8400096C          A    L  ONE
0A50 00 D6800000          STO  I2
0A52 0  D200              STO   2
0A53 00 4C800A4B          B    I  INC
0A55 0  0A56        INTER DC      *
0A56 0  0000              DC          INTERPRET
0A57 00 C4000AD8          LD   L  E
0A59 00 D4800B06          STO  I  R
0A5B 00 6F000AD8          STX  L3 E
0A5D 00 74010B06          MDM  L  R,1
0A5F 0  C1FD              LD    1 -3   C
0A60 00 D4800B06          STO  I  R
0A62 00 74010B06          MDM  L  R,1
0A64 0  C003              LD    3 3
0A65 0  D1FD              STO   1 -3   C
0A66 0  C1FE              LD    1 -2   ACCEPT
0A67 00 D4800B06          STO  I  R
0A69 00 74010B06          MDM  L  R,1
0A6B 0  C003              LD      ARETR
0A6C 0  D1FE              STO   1 -2   ACCEP
0A6D 00 4C800A56          B    I  INTER+1
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PAGE   6

0A6F 0  091D        ARETR DC      RETRV
0A70 0  0000        COM   DC        END INTERPRET
0A71 00 74FD0B06          MDM  L  R,-3
0A73 00 67800B06          LDX  I3 R
0A75 0  C301              LD    3 1
0A76 0  D1FD              STO   1 -3   C
0A77 0  C302              LD    3 2
0A78 0  D1FE              STO   1,-2   ACCEPT
0A79 0  C300              LD    3
0A7A 00 94000AD8          S     L  E
0A7C 00 4C880A70          BNP   I  COM
0A7E 0  C300              LD    3 0
0A7F 00 D4000AD8          STO   L  E
0A81 00 4C800A70          B     I  COM
0A83 0  0000        LOC   DC          LOCATION OF CODE
0A84 00 44000922          BSI   L  NEXT
0A86 00 44000993          BSI   L  DO
0A88 0  C302              LD    3 2
0A89 0  7201              MDX   2 1
0A8A 0  D200              STO   2
0A8B 00 4C800A83          B     I  LOC
0A8D 0  0000        OR    DC        OR TOP OF STACK
0A8E 0  C200              LD    2
0A8F 0  72FF              MDX   2 -1
0A90 0  EA00              OR    2
0A91 0  D200              STO   2
0A92 00 4C800A8D          B     I  OR
0A94 0  0000        STORE DC        STORE TO T.O.S
0A95 0  C2FF              LD    2 -1
0A96 00 D6800000          STO   I2
0A98 0  72FE              MDX   2 -2
0A99 00 4C800A94          B     I  STORE
0A9B 0  0000        SD    DC        STACK TO DEPOSIT
0A9C 0  C200              LD    2
0A9D 00 74010D83          MDM   L  IC,1
0A9F 00 D4800D83          STO   I  IC
0AA1 0  72FF              MDX   2 -1
0AA2 00 4C800A9B          B     I  SD
0AA4 0  0AA5        ADDR  DC        *
0AA5 0  0000              DC        PLACE ADDRESS ON STACK
0AA6 0  7201              MDX   2 1
0AA7 00 6F00096D          STX   L3 TEMP
0AA9 00 7403096D          MDM   L  TEMP,3
0AAB 00 C400096D          LD    L  TEMP
0AAD 0  D200              STO   2
0AAE 00 4C800AA5          B     I  ADDR+1
0AB0 0  0AB1        LITER DC        *
0AB1 0  0000              DC        PLACE VALUE ON STACK
0AB2 0  7201              MDX   2 1
0AB3 0  C303              LD    3 3
0AB4 0  D200              STO   2
0AB5 00 4C800AB1          B     I  LITER+1
0AB7 0  0000        OPER  DC        OPERATION
0AB8 00 440009C1          BSI   L  ENTRY
0ABA 00 74010D83          MDM   L  IC,1
0ABC 00 C4000D83          LD    L  IC
0ABE 0  7201              MDX   2 1
0ABF 0  D200              STO   2
0AC0 0  D302              STO   3 2
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PAGE   7

0AC1 00 4C800AB7          B    I  OPER
0AC3 0  0000       CONS   DC         CONSTANT
0AC4 00 44000922          BSI  L  NEXT
0AC6 00 440009C1          BSI  L  ENTRY
0AC8 0  C0E7              LD      LITER
0AC9 0  D302              STO   3 2
0ACA 0  C200              LD    2
0ACB 0  D303              STO   3 3
0ACC 0  72FF              MDX   2 -1
0ACD 00 4C800AC3          B    I  CONS
0ACF 0  0000       INTEG  DC         DECLARE INTEGER
0AD0 00 67800AD8          LDX  I3 E
0AD2 0  C0D1              LD      ADDR
0AD3 0  D302              STO   3 2
0AD4 0  40D0              BSI     ADDR+1
0AD5 00 4C800ACF          B    I  INTEG
0AD7 0  0000       E1     DC         TOP OR SYMBOL TABLE
0AD8 0  0000       E      DC         THE PLACE TO START SEARCHES
0AD9 0  15C4       W1     DC       2*WORD   -6
0ADA 0  0000              DC         D   -5   SAVE CHARACTER
0ADB 0  0000              DC         -4   SAVE OPERATION
0ADC 0  0000              DC         C   -3   CURRENT CHARACTER
0ADD 0  0000              DC         -2   ACCEPT
0ADE 0  0000              DC         -1   CHARACTER TABLE
0ADF 0  0000       A      DC         A   XR1   CURRENT CHARACTER
0AE0 0  0000       N      DC         1
0AE1 0  0000              DC         W   2   WORD CHARACTER
0AE2    0014       WORD   BSS      20   3
0AF6    0010       STACK  BSS      16
0B06 0  0000       R      DC
0B07    0020       INTST  BSS      32
0B27 0  1AE0       C1     DC       2*SECT+642+72   RESET CHARACTER
0B28 0  0000       C2     DC         CHARACTER BEYOND RECORD
0B29 0  1860       C3     DC       2*SECT+74   VHARACTER BEYOND SECTOR
0B2A 0  0000       ERROR  DC       0
0B2B 00 4C800B2A          B    I  ERROR
0B2D 0  0000       RECOR  DC       0   NEXT RECORD
0B2E 0  C1FD              LD    1 -3   C
0B2F 0  90F9              S       C3
0B30 00 44180B3B          BSI  L  BLOCK,+-   NEXT BLOCK
0B32 0  C1FD              LD    1 -3   C
0B33 0  9006              S       D152
0B34 0  D1FD              STO   1 -3
0B35 00 74B00B28          MDM  L  C2,-80
0B37 0  4017              BSI     FIXUP
0B38 00 4C800B2D          B    I  RECOR
0B3A 0  0098       D152   DC       152
0B3B 0  0000       BLOCK  DC       0   NEXT BLOCK
0B3C 00 678009F9          LDX  I3 X3
0B3E 20 042624B1          LIBF    DISK1
0B3F 0  1000              DC      /1000
0B40 0  0C0A              DC      BUF
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PAGE    8

0B41 0  0B2A                 DC       ERROR
0B42 20 042624B1    BL1      LIBF     DISK1
0B43 0  0000                 DC       /0000
0B44 0  0C0A                 DC       BUF
0B45 0  70FC                 B        BL1
0B46 00 74010C0B             MDM   L  SECT,1
0B48 00 C4000B27             LD    L  C1
0B4A 0  D1FD                 STO    1 -3   C
0B4B 00 D4000B28             STO   L  C2
0B4D 00 4C800B3B             B     I  BLOCK
0B4F 0  0000        FIXUP    DC          RE-ARRANGE RECORD
0B50 0  C1FD                 LD     1 -3   C
0B51 0  1801                 SRA      1
0B52 00 D4000B6D             STO   L  FX1
0B54 00 D4000B6E             STO   L  FX2
0B56 00 74270B6E             MDM   L  FX2,39
0B58 0  6314                 LDX    3 20
0B59 00 C4800B6D    FI1      LD    I  FX1
0B5B 00 D400096D             STO   L  TEMP
0B5D 00 C4800B6E             LD    I  FX2
0B5F 00 D4800B6D             STO   I  FX1
0B61 00 C400096D             LD    L  TEMP
0B63 00 D4800B6E             STO   I  FX2
0B65 00 74010B6D             MDM   L  FX1,1
0B67 00 74FF0B6E             MDM   L  FX2,-1
0B69 0  73FF                 MDX    3 -1
0B6A 0  70EE                 B        FI1
0B6B 00 4C800B4F             B     I  FIXUP
0B6D 0  0000        FX1      DC
0B6E 0  0000        FX2      DC
0B6F 0  0000        PUT      DC          OUTPUT CHARACTER
0B70 0  C100                 LD     1     A
0B71 0  D02F                 STO       AP
0B72 00 67800BA1             LDX   I3  AP
0B74 00 C7000BCA             LD    L3  BCD
0B76 0  D100                 STO    1     A
0B77 00 C4000BA3             LD    L  DP
0B79 00 4400097F             BSI   L  DEPOS
0B7B 00 74010BA3             MDM   L  DP,1
0B7D 0  C025                 LD       DP
0B7E 0  9025                 S        DP1
0B7F 00 44180B85             BSI   L  PRINT,+-
0B81 0  C01F                 LD       AP
0B82 0  D100                 STO    1     A
0B83 00 4C800B6F             B     I  PUT
0B85 0  0000        PRINT    DC       0   PRINT RECORD
0B86 00 C4000C09             LD    L  BCDBL
0B88 0  D100                 STO    1     A
0B89 0  C019        PR2      LD       DP
0B8A 0  9019                 S        DP1
0B8B 00 4C180B94             BZ       PR3
0B8D 00 C4000BA3             LD    L  DP
0B8F 00 4400097F             BSI   L  DEPOS
0B91 00 74010BA3             MDM   L  DP,1
0B93 0  70F5                 B        PR2
0B94 00 678009F9    PR3      LDX   I3  X3
0B96 20 176558F1             LIBF     PRNT1
0B97 0  2000                 DC       /2000
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PAGE    9

0B98 0  0BA5                    DC      PRN
0B99 0  0B2A                    DC      ERROR
0B9A 20 176558F1    PR1         LIBF    PRNT1
0B9B 0  0000                    DC      /0000
0B9C 0  70FD                    B       PR1
0B9D 0  C004                    LD      DP0
0B9E 0  D004                    STO     DP
0B9F 00 4C800B85                B     I PRINT
0BA1 0  0000        AP          DC
0BA2 0  174C        DP0         DC      2*PRN+2
0BA3 0  174C        DP          DC      2*PRN+2
0BA4 0  1794        DP1         DC      2*PRN+74
0BA5 0  0024        PRN         DC      36
0BA6    0024                    BSS     36
0BCA 0  00F0        BCD         DC      240   0
0BCB 0  00F1                    DC      241   1
0BCC 0  00F2                    DC      242   2
0BCD 0  00F3                    DC      243   3
0BCE 0  00F4                    DC      244   4
0BCF 0  00F5                    DC      245   5
0BD0 0  00F6                    DC      246   6
0BD1 0  00F7                    DC      247   7
0BD2 0  00F8                    DC      248   8
0BD3 0  00F9                    DC      249   9
0BD4 0  00C1                    DC      193   A
0BD5 0  00C2                    DC      194   B
0BD6 0  00C3                    DC      195   C
0BD7 0  00C4                    DC      196   D
0BD8 0  00C5                    DC      197   E
0BD9 0  00C6                    DC      198   F
0BDA 0  00C7                    DC      199   G
0BDB 0  00C8                    DC      200   H
0BDC 0  00C9                    DC      201   I     /12
0BDD 0  00D1                    DC      209   J
0BDE 0  00D2                    DC      210   K
0BDF 0  00D3                    DC      211   L
0BE0 0  00D4                    DC      212   M
0BE1 0  00D5                    DC      213   N
0BE2 0  00D6                    DC      214   O     /18
0BE3 0  00D7                    DC      215   P
0BE4 0  00D8                    DC      216   Q
0BE5 0  00D9                    DC      217   R
0BE6 0  00E2                    DC      226   S     /1C
0BE7 0  00E3                    DC      227   T
0BE8 0  00E4                    DC      228   U
0BE9 0  00E5                    DC      229   V
0BEA 0  00E6                    DC      230   W
0BEB 0  00E7                    DC      231   X
0BEC 0  00E8                    DC      232   Y
0BED 0  00E9                    DC      233   Z
0BEE 0  0040                    DC      64    BLANK   /24
0BEF 0  004A                    DC      74    CENTS
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PAGE  10

0BF0 0  007B              DC      123   NUMBER
0BF1 0  004C              DC      76    LESS THAN
0BF2 0  004D              DC      77    (
0BF3 0  004E              DC      78    +
0BF4 0  004F              DC      79    STROKE
0BF5 0  0050              DC      80    AMPERSAND   /2B
0BF6 0  005A              DC      90    EXCLAME
0BF7 0  005B              DC      91    $
0BF8 0  005C              DC      92    *
0BF9 0  005D              DC      93    )
0BFA 0  005E              DC      94    SEMI COLON
0BFB 0  005F              DC      95    NOT
0BFC 0  0060              DC      96    -
0BFD 0  0061              DC      97    /
0BFE 0  006B              DC      107   ,   /34
0BFF 0  006C              DC      108   PERCENT
0C00 0  006D              DC      109   UNDERSCORE
0C01 0  006E              DC      110   GREATER THAN
0C02 0  006F              DC      111   QUESTION
0C03 0  007A              DC      122   COLON
0C04 0  004B              DC      75    .   3D
0C05 0  007C              DC      124   AT
0C06 0  007D              DC      125   '
0C07 0  007E              DC      126   =
0C08 0  007F              DC      127   QUOTE   /3E
0C09 0  0040        BCDBL DC      64    BL
0C0A 0  0140        BUF   DC      320   SECTOR BUFFER
0C0B 0  0000        SECT  DC
0C0C    0140              BSS     320
0D4C 0  0000              DC
0D4D 0  0000              DC
0D4E 0  09A3              DC      UNDEF
0D4F 0  0000              DC
0D50 0  0F18              DC      /0F18   FORTH
0D51 0  1B1D              DC      /1B1D
0D52 0  0A31              DC      FORTH
0D53 0  0000              DC
0D54 0  1B0E              DC      /1B0E   RECURSE
0D55 0  0C1E              DC      /0C1E
0D56 0  0AB1              DC      LITER+1
0D57 0  0B06              DC      R
0D58 0  0F12              DC      /0F12   FIND
0D59 0  170D              DC      /170D
0D5A 0  0993              DC      DO
0D5B 0  0000              DC
0D5C 0  0A0D              DC      /0A0D   ADDRESS
0D5D 0  0D1B              DC      /0D1B
0D5E 0  0AA5              DC      ADDR+1
0D5F 0  0000              DC
0D60 0  0E17              DC      /0E17   END
0D61 0  0D24              DC      /0D24
0D62 0  0A70              DC      COM
0D63 0  0000              DC
0D64 0  110E              DC      /110E   HEX
0D65 0  2124              DC      /2124
0D66 0  09AB              DC      HEX
0D67 0  0000              DC
0D68 0  181B              DC      /181B   OR
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PAGE  11

0D69  0  2424              DC      /2424
0D6A  0  0A8D              DC      OR
0D6B  0  0000              DC
0D6C  0  3D24              DC      /3D24    =
0D6D  0  2424              DC      /2424
0D6E  0  0A94              DC      STORE
0D6F  0  0000              DC
0D70  0  3924              DC      /3924    COLON
0D71  0  2424              DC      /2424
0D72  0  09D2              DC      ENTER
0D73  0  0000              DC
0D74  0  3A24              DC      /3A24    .
0D75  0  2424              DC      /2424
0D76  0  09D2              DC      ENTER
0D77  0  0000              DC
0D78  0  3024              DC      /3024    SEMICOLON
0D79  0  2424              DC      /2424
0D7A  0  0A70              DC      COM
0D7B  0  0000              DC
0D7C  0  3424              DC      /3424    ,
0D7D  0  2424              DC      /2424
0D7E  0  0A70              DC      COM
0D7F  0  0000              DC
0D80  0  120C              DC      /120C    IC
0D81  0  2424              DC      /2424
0D82  0  0AA5              DC      ADDR+1
0D83  0  0000        IC    DC
0D84  0  2524              DC      /2524    CENT
0D85  0  2424              DC      /2424
0D86  0  0AB7              DC      OPER
0D87  0  0000              DC
0D88  0  1819              DC      /1819    OPERATION
0D89  0  0E1B              DC      /0E1B
0D8A  0  0AB7              DC      OPER
0D8B  0  0000              DC
0D8C  0  0E17              DC      /0E17    ENTRY
0D8D  0  1D1B              DC      /1D1B
0D8E  0  09C1              DC      ENTRY
0D8F  0  0000              DC
0D90  0  1217              DC      /1217    INTEGER
0D91  0  1D0E              DC      /1D0E
0D92  0  0ACF              DC      INTEG
0D93  0  0000              DC
0D94  0  1217              DC      /1217    INC
0D95  0  0C24              DC      /0C24
0D96  0  0A4B              DC      INC
0D97  0  0000              DC
0D98  0  1C0D              DC      /1C0D    SD
0D99  0  2424              DC      /2424
0D9A  0  0A9B              DC      SD
0D9B  0  0000              DC
0D9C  0  0C18              DC      /0C18    CONVERT
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PAGE  12

0D9D 0  171F                DC      /171F
0D9E 0  0900                DC      CONVE
0D9F 0  0000                DC
0DA0 0  0F0E                DC      /0F0E   FETCH
0DA1 0  1D0C                DC      /1D0C
0DA2 0  096E                DC      FETCH
0DA3 0  0000                DC
0DA4 0  0D0E                DC      /0D0E   DEPOSIT
0DA5 0  1918                DC      /1918
0DA6 0  097F                DC      DEPOS
0DA7 0  0000                DC
0DA8 0  191E                DC      /191E   PUT
0DA9 0  1D24                DC      /1D24
0DAA 0  0B6F                DC      PUT
0DAB 0  0000                DC
0DAC 0  191B                DC      /191B   PRINT
0DAD 0  1217                DC      /1217
0DAE 0  0B85                DC      PRINT
0DAF 0  0000                DC
0DB0 0  170E                DC      /170E   NEXT
0DB1 0  211D                DC      /211D
0DB2 0  0922                DC      NEXT
0DB3 0  0000                DC
0DB4 0  1518                DC      /1518   LOC
0DB5 0  0C24                DC      /0C24
0DB6 0  0A83                DC      LOC
0DB7 0  0000                DC
0DB8 0  0E24                DC      /0E24   E
0DB9 0  2424                DC      /2424
0DBA 0  0AB1                DC      LITER+1
0DBB 0  0AD8                DC      E
0DBC 0  1512                DC      /1512   LIT
0DBD 0  1D24                DC      /1D24
0DBE 0  0AB1                DC      LITER+1
0DBF 0  0000                DC
0DC0 0  0E01        E2      DC      /0E01   E1
0DC1 0  2424                DC      /2424
0DC2 0  0AB1                DC      LITER+1
0DC3 0  0AD7                DC      E1
0DC4    09FA                END     START-1

   000 OVERFLOW SECTORS SPECIFIED
   000 OVERFLOW SECTORS REQUIRED
   099 SYMBOLS DEFINED
    NO ERROR(S)  FLAGGED IN ABOVE ASSEMBLY

// DUP

*DELETE              FORTX
CART ID 0001   DB ADDR  3700   DB CNT   0060

*STORECI1   WS  UA  FORTX
R 41  3090 (HEX) WDS UNUSED BY CORE LOAD
LIBF TRANSFER VECTOR
 PRNT1  0DC4
 DISK1  00F9
SYSTEM SUBROUTINES
 ILS04  00C4
```